Use the Unit Circle to Define Trigonometric Functions

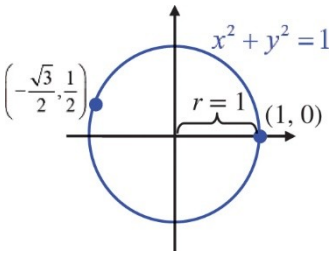
Unit Circle: The circle of radius 1 unit, centered at the origin.

$$x^2 + y^2 = 1$$

Each point (x, y) on the unit circle satisfies the equation .

1. Show that the point $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ lies on the unit circle.

$$x^2 + y^2 = 1$$
$$\left(-\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = 1$$
$$\frac{3}{4} + \frac{1}{4} = 1 \quad \surd$$



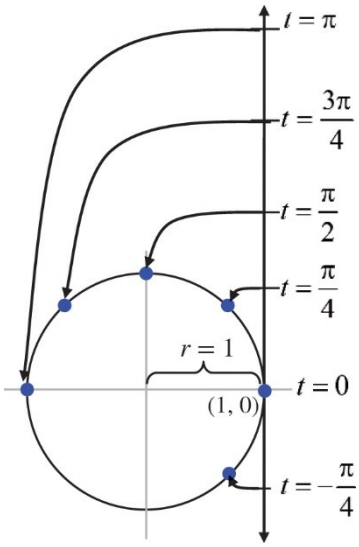
Suppose we wrap the real number line around the unit circle by placing the point $t = 0$ from the number line on the point (1, 0) on the circle.

Positive real numbers from the number line ($t > 0$) wrap onto the unit circle in the counterclockwise direction.

Negative numbers from the number line ($t < 0$) wrap onto the unit circle in the clockwise direction.

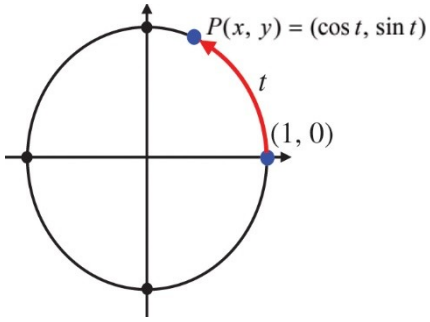
This associates each real number t with a unique point (x, y) on the unit circle.

This enables us to define trigonometric functions as *circular functions*.



Unit Circle Definition of Trigonometric Functions

Let $P(x, y)$ be the point associated with a real number t measured along the circumference of the unit circle from the point (1, 0).



$$\sin t = y \qquad \csc t = \frac{1}{y} \quad (y \neq 0)$$

$$\cos t = x \qquad \sec t = \frac{1}{x} \quad (x \neq 0)$$

$$\tan t = \frac{y}{x} \quad (x \neq 0) \qquad \cot t = \frac{x}{y} \quad (y \neq 0)$$

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If P is on the y-axis [either (0, 1) or (0, −1)], then $x = 0$ and the tangent and secant functions are undefined.

If P is on the x-axis [either (1, 0) or (−1, 0)], then $y = 0$ and the cotangent and cosecant functions are undefined.

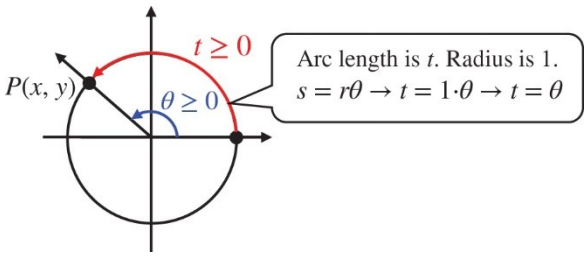
2. Suppose that the real number t corresponds to the point $P\left(\frac{5}{13}, -\frac{12}{13}\right)$ on the unit circle. Evaluate $\sin t$, $\cos t$, and $\tan t$.

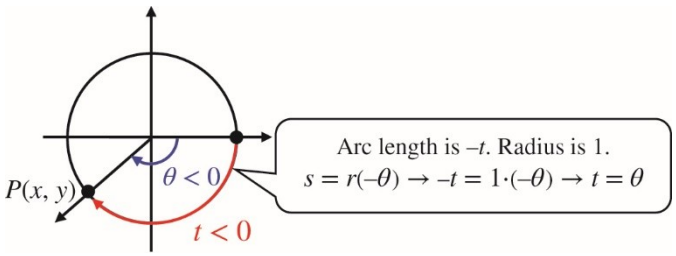
$\sin t = y = -\frac{12}{13}$ | $\cos t = x = \frac{5}{13}$ | $\tan t = \frac{y}{x} = \frac{-\frac{12}{13}}{\frac{5}{13}} = -\frac{12}{5}$

Evaluate Trigonometric Functions Using the Unit Circle

Let $P(x, y)$ be the point on the unit circle associated with a real number t. Let θ be the central angle in standard position measured in radians with terminal side through P and arc length t.

Because the radius of the unit circle is 1, the arc length formula $s = r\theta$ becomes $t = 1 \cdot \theta$ or simply $t = \theta$.





Trigonometric Functions of Real Numbers and Angles

If $\theta = t$ radians, then

$\sin t = \sin \theta$ $\quad \cos t = \cos \theta$ $\quad \tan t = \tan \theta$

$\csc t = \csc \theta$ $\quad \sec t = \sec \theta$ $\quad \cot t = \cot \theta$

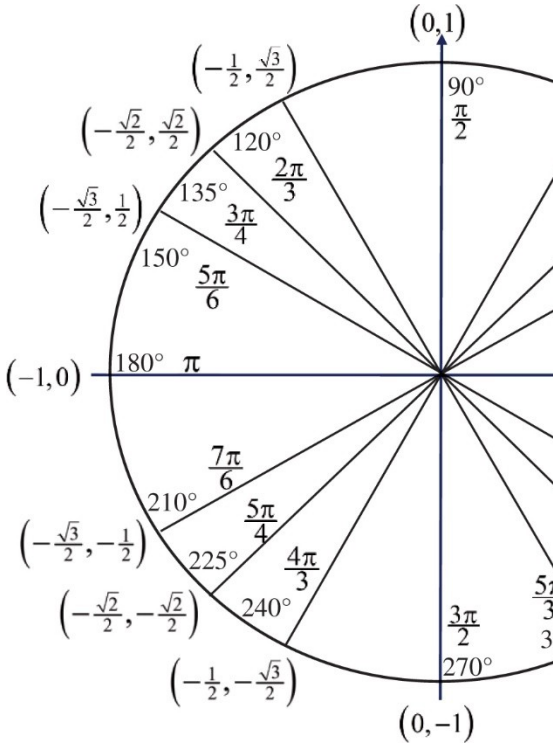
The trigonometric functions of an angle θ can now be evaluated using the unit circle.

Angle θ can be measured in radians or degrees because the values of the functions depend on the *coordinates* of $P(x, y)$ where the terminal side of θ intercepts the unit circle.

As a result, the angle may be measured in radians or degrees.

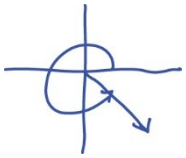
Note the symmetry of the unit circle.

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3. Identify the ordered pair on the unit circle that corresponds to the number $t = \frac{5\pi}{3}$. Then evaluate sin t and sec t.



ordered pair $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

$\sin t = -\frac{\sqrt{3}}{2}$

$\sec t = \frac{1}{\cos t} = \frac{1}{\frac{1}{2}} = 2$

4. Evaluate cos t and tan t for $t = -\frac{5\pi}{6}$. $(x, y) = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

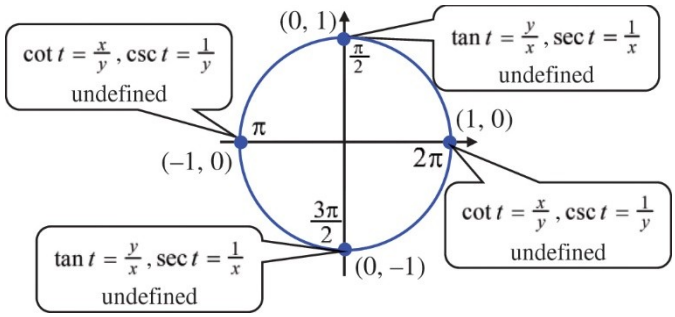
$\cos t = \cos\left(-\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

$\tan t = \frac{y}{x} = \frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

Identify the Domains of the Trigonometric Functions

$\sin t = y$ $\cos t = x$

and have no restrictions on their domain.



$\tan t = \frac{y}{x}$ $\sec t = \frac{1}{x}$

and are undefined for all values of t corresponding to the points (0, 1) and (0, −1) on the unit circle.

$\cot t = \frac{x}{y}$ $\csc t = \frac{1}{y}$

and are undefined for all values of t corresponding to the points (1, 0) and (−1, 0) on the unit circle.

Function	Domain	Notes
$f(t) = \sin t$, $f(t) = \cos t$	All real numbers	No restrictions

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$f(t) = \tan t$, $f(t) = \sec t$	$\left\{ t \mid t \neq \frac{(2n+1)\pi}{2} \text{ for all integers } n \right\}$	Exclude odd multiples of $\frac{\pi}{2}$ (90°).
$f(t) = \cot t$, $f(t) = \csc t$	$\{ t \mid t \neq n\pi \text{ for all integers } n \}$	Exclude multiples of π (180°).

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Apply the Period and Properties of Trigonometric Functions

Periodic Function: A function f is **periodic** if $f(t+p)=f(t)$ for some constant p.
(f repeats at regular intervals)
The smallest positive value p for which f is periodic is called the **period** of f.

Sine, cosine, cosecant, and secant are periodic with period 2π.
Tangent and cotangent are periodic with period π.

Periodic Properties of Trigonometric Functions			
Function	**Period**	**Property**	
Sine	2π (360°)	$\sin(t+2\pi)=\sin t$	$\sin(\theta+360°)=\sin\theta$
Cosine	2π (360°)	$\cos(t+2\pi)=\cos t$	$\cos(\theta+360°)=\cos\theta$
Cosecant	2π (360°)	$\csc(t+2\pi)=\csc t$	$\csc(\theta+360°)=\csc\theta$
Secant	2π (360°)	$\sec(t+2\pi)=\sec t$	$\sec(\theta+360°)=\sec\theta$
Tangent	π (180°)	$\tan(t+\pi)=\tan t$	$\tan(\theta+180°)=\tan\theta$
Cotangent	π (180°)	$\cot(t+\pi)=\cot t$	$\cot(\theta+180°)=\cot\theta$

Use the unit circle and the period of the function to evaluate the expression if possible.

5. $\cos(-315°)=\cos 45°=\dfrac{\sqrt{2}}{2}$

$360°-315°=45°$

6. $\tan\left(-\dfrac{\pi}{2}\right)=\tan\dfrac{\pi}{2}\rightarrow$ undefined

$-\dfrac{\pi}{2}+\pi=\dfrac{\pi}{2}$

7. $\sin\dfrac{17\pi}{6}=\sin\dfrac{5\pi}{6}=\dfrac{1}{2}$

$\dfrac{17\pi}{6}-\dfrac{12\pi}{6}=\dfrac{5\pi}{6}$

(2π)

8. $\sec 510°=\sec 150°=-\dfrac{2}{\sqrt{3}}$ or $-\dfrac{2\sqrt{3}}{3}$ ($\dfrac{1}{\cos 150°}$)

$510°-360°=150°$

9. $\cot\dfrac{7\pi}{6}=\cot\dfrac{\pi}{6}=\sqrt{3}$

$\dfrac{7\pi}{6}-\dfrac{6\pi}{6}=\dfrac{\pi}{6}$

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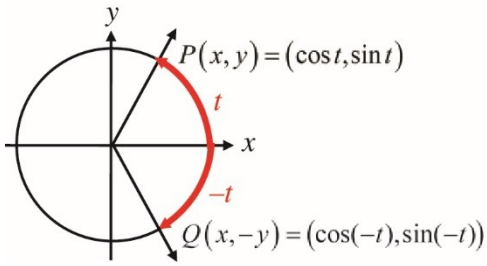
10. $\csc 540° = \csc 180°$ undefined $540° - 360° = 180°$

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A function f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$.

$\cos(-t) = \cos t = x$

The cosine function is an even function.



$\sin t = y$ but $\sin(-t) = -y$. Thus, $\sin(-t) = -\sin t$.

The sine function is an odd function.

$\tan(-t) = \dfrac{-y}{x}$ and $\tan t = \dfrac{y}{x}$. Therefore $\tan(-t) = -\tan t$.

The tangent function is also an odd function.

Summary: cos t and sec t are even functions. sin t, csc t, tan t, and cot t are odd functions.

Use the properties of the trigonometric functions to simplify.

11. $\tan(t+\pi)\cdot\cos(t+2\pi)$

$= \tan t \cdot \cos t$

$= \dfrac{\sin t}{\cos t} \cdot \cos t$

$= \sin t$

12. $3\sin t + \sin(2\pi - t)$

$= 3\sin t + \sin(-t)$

$= 3\sin t - \sin t$

$= 2\sin t$

13. $\sin(-t)\cdot\sec t - \tan(-t) = -\sin t \cdot \sec t - (-\tan t)$

$= -\sin t \cdot \dfrac{1}{\cos t} + \dfrac{\sin t}{\cos t}$

$= -\dfrac{\sin t}{\cos t} + \dfrac{\sin t}{\cos t}$

$= 0$

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